Padraig Lavery

Software Engineer – Security, Infrastructure & Reliability

Belfast, Northern Ireland | LinkedIn | GitHub

PROFILE

Software engineer with 4+ years building and securing high-volume banking systems for UK and Middle East clients on AWS and Kubernetes. Has delivered PCI DSS vulnerability remediation, audit logging and service hardening, and writes the tooling and guides that make the secure path the easy one for other engineers. Cares about reliability and security in production, not only shipping features.

EXPERIENCE

Developer | Reward, Belfast | June 2022 – Present

Security engineering

- Designed and implemented audit logging of staff access to customer accounts across UK and Middle East bank portals (agent, customer, timestamp, session, IP) to meet client security and compliance audits.
- Hardened authentication and account services: removed stack traces and SQL details from error responses and standardised input validation across partner microservices so malformed requests no longer trigger database errors.
- Remediated critical Snyk-flagged vulnerabilities for PCI DSS compliance as part of planned AWS SDK v3 and Node 22 Lambda runtime upgrades across repositories and bank deployments.
- Audited 100+ S3 buckets, applying lifecycle policies and removing obsolete resources to reduce compliance risk and storage cost.

Tooling and developer experience

- Authored the engineering migration guide for AWS SDK for JavaScript v2 to v3 across Lambdas, Fargate services and microservices.
- Led a cross-team initiative for automated API testing: request validation, shared error handling, safe test-data cleanup endpoints, OpenAPI documentation, unit tests, and stable GitHub Actions CI.
- Authored runbooks, SOPs and technical guides for the team and for production services.

Reliability in distributed systems

- Eliminated recurring daily out-of-memory crashes in a transaction ingestion service by loading caches once per batch instead of per file.
- Redesigned a month-end invoicing batch with per-partner error isolation, explicit timeouts and safe retry/reconciliation.
- Resolved a customer-facing outage in a Kubernetes/Istio ingress setup and corrected the Helm templates behind it.

Intern Software Engineer | Flexera, Belfast | May 2020 – April 2021

- Built a cloud-cost API with Go, Docker and Kubernetes.

SKILLS

- **Security:** PCI DSS remediation, Snyk, OWASP, audit logging, input validation, secure error handling
- **Cloud and infrastructure:** AWS (Lambda, Fargate, S3, EventBridge), Kubernetes, Helm, Istio, Docker, Terraform, GitHub Actions
- **Languages:** JavaScript (Node.js), Python, C#, Go, SQL
- **Back end and data:** Node.js, .NET Core, FastAPI, microservices, OpenAPI, PostgreSQL, MongoDB

PROJECTS

AI-Based Credit Card Fraud Detection (final year project) | Angular, Flask, GraphQL, Docker, Kubernetes, AWS SageMaker

- End-to-end web app flagging irregular card transactions, run as Docker/Kubernetes microservices on AWS.

Incident Copilot (in progress) | Python, FastAPI, PostgreSQL + pgvector, Ollama

- Incident-triage service that retrieves runbook content and suggests likely causes and next steps.

EDUCATION AND CERTIFICATIONS

- BSc (Hons) Computer Science, 2:1 – Ulster University, Belfast, June 2022

- AWS Certified Developer – Associate; AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner – preparing, exam November 2026